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ITBDA4-14 PROJECT 2

This project applies big data analytics techniques to text data in four tasks: a MapReduce keyword search in R using Hadoop Streaming; LDA topic modelling of a news corpus; a web text API prototype that scrapes, preprocesses and summarises pages using TextRank and frequency-based scoring; and an ETL workflow that cleans scraped web data in R, loads it into PostgreSQL and analyses it with SQL. The work is published in a GitHub repository for reproducibility.

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Question 1

1.1

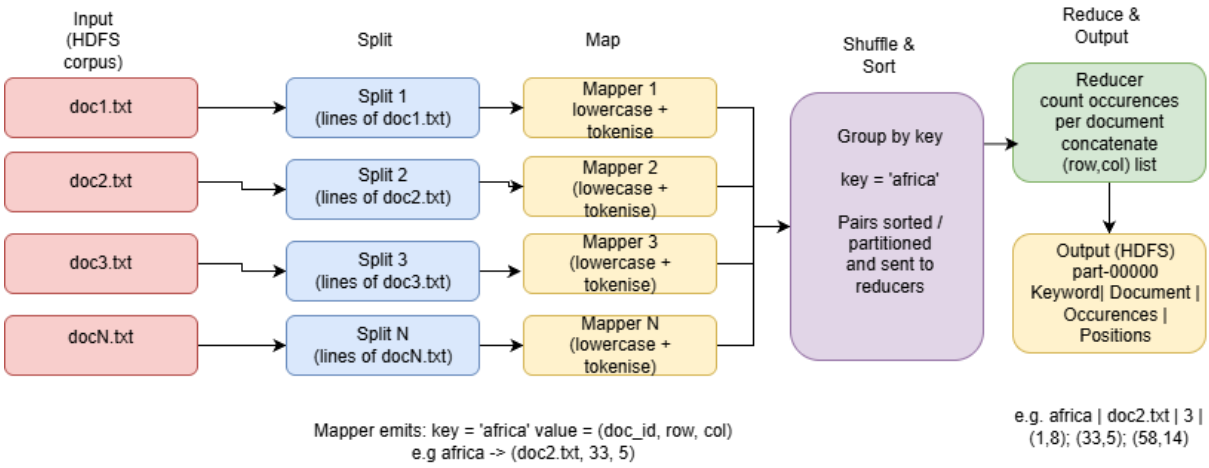


Figure 1: Logical Layout of the MapReduce keyword search

- Input: The corpus (map_reduce_test_docs.zip) is made available as plain text documents (doc1.txt, doc2.txt and so on) in the distributed file system (HDFS) , where large files are stored in replicated blocks across data nodes (Mazumdar, S., Seybold, D., Kritikos, K. and Verginadis, Y., 2019).
- Split: The framework divides the input into splits and assigns each split to a map task, ideally on the node that already stores the data, which reduces network traffic.
- Map: each Mapper reads its split line by line, keeping the document name and the line number (the row). Each line is converted to lower case and tokenised into words on punctuation and white space. The position of each word within the line is the column. Whenever a token = ‘africa’, the Mapper emits the pair key = ‘africa’ , value = ‘AFRICA’ and ‘africa’ equivalent, which satisfies the case insensitive requirement.
- Shuffle & Sort: The framework collects pairs emitted by all Mappers, groups them by key and delivers each group to a Reducer, sorted. No user code is needed in this phase.
- Reduce: The Reducer groups the values by document, counts the occurrences per document and concatenates the (row, column) coordinates into a position list.
- Output: The result is written back to HDFS as one record per document: keyword, document, number of occurrences and the list of (row, column) positions. (Zhang, 2018)

1.2 The mapper works is implemented as the function run_mapper(). It unzips the corpus, reads every document line by line, tokenises each line and emits one

intermediate record for every occurrence of the keyword. The record has four fields that correspond to the key value pair (Zhang, 2018).

The mapper was executed on the corpus and its output (the intermediate key value records) was passed to the Reducer.

1.3 The Reducer is implemented as run_reducer(). It receives the intermediate records from the Mapper, groups them by key and document, counts the records in each group and joins the coordinates into a single text field.

```
 QUESTION 1.2 & 1.3: MAPREDUCE REDUCED OUTPUT
> print(q1_final_result)
# A tibble: 4 × 4
  Keyword Document Occurrences Position_List
  <chr>   <chr>          <int> <chr>
1 africa  doc1.txt           1 (1,11)
2 africa  doc2.txt           3 (1,8); (33,5); (58,14)
3 africa  doc3.txt           8 (1,8); (3,2); (5,2); (40,22); (46,2); (50,23); (52,15); (54,6)
4 africa  doc5.txt           4 (1,12); (3,14); (23,21); (38,6)
>
```

Figure 2: Console Output of print(q1_final_result)

Interpretation

The keyword “africa” occurs 16 times in total across four documents (1 + 3 + 8 + 4). Document doc3.txt has the most occurrences (8) which is half of all matches and it uses the word on lines 1, 3, 5, 40, 46, 50, 52 and 54. This suggests that doc3.txt is the article most focused on Africa. The keyword appears on line 1 of doc2, doc3 and doc5, which is where a news headline would be expected and this is a further indication that these articles are about the continent rather than mentioning it in passing. Document doc1.txt contains a single occurrence, again on line 1 so the keyword appears in its headline only.

1.4 The job is submitted to the cluster through Hadoop Streaming, which lets any executable that reads from standard input and writes to standard output act as the Mapper and the Reducer, the framework still performs splitting , shuffling and sorting between the two (Ginde, G., Aedula, R., Saha, S., Mathur, A., Dey, S.R., Sampatrao, G.S. and Sagar, B.D., 2017). Since Hadoop is not installed, the function reports below (Figure 3) and job itself was run using Mapper and Reducer.

```
QUESTION 1.4: HADOOP STREAMING CONFIGURATION
Hadoop binary not found in system PATH. Configuration payload verified successfully for cluster
execution.
```

Figure 3:Hadoop Streaming Configuration Result

Question 2

2.1 The archive news.zip was unzipped into a temporary folder and every file was read with readLines(). The lines of each article were collapsed into one character string with paste (collapse = “ ”), so that each article occupies exactly one row. The

result is the data frame news_df with two colums : file_name (name of the source file) and content (full text of the article).

Below shows the ten largest document term counts after denoising. The doc_id is the row position of the article in news_df, not necessarily the number in its file name.

```
>
> cat("2.1 DENOISED TOKENIZED SAMPLE\n")
2.1 DENOISED TOKENIZED SAMPLE
> print(head(word_counts, 10))
   doc_id         word  n
1       7     invoices 17
2       9   government 14
3       9    coalition 13
4       7         days 11
5       1        eskom 10
6       7  departments 10
7       8       public  9
8       2         cape  8
9       4 unemployment  8
10      7         paid  8
>
```

Figure 4: Result after denoising

Interpretation

No stop words, numbers or symbols appear among the most frequent terms, which shows that the denoising steps worked as intended: the remaining high count words are all content words that describe the subject of an article. The articles are clearly not about a single subject. Document 7 is dominated by invoice-payment vocabulary (invoices, days, departments, paid), document 9 by government and coalition, document 1 by Eskom, document 4 by unemployment and document 2 by a place name. Most of this vocabulary relates to government and public sector matters, which is the kind of structure that the topic model should recover. One limitation is visible already : no stemming or lemmatisation was applied so the singular “department” and the plural “departments” are counted as different terms.

2.2 LDA is a generative probabilistic model in which each document is a mixture of a small number of topics and each topic is a probability distribution over the vocabulary (Denny, 2018). Beta is chosen and produces the top words per topic :

```
 2.2 TOP K WORDS PER TOPIC (BETA PROBABILITIES)
> print(q2_top_terms)
# A tibble: 10 × 3
   topic term             beta
   <int> <chr>           <dbl>
 1     1 public         0.0239
 2     1 government     0.0184
 3     1 department     0.0157
 4     1 coalition      0.0120
 5     1 administration 0.0102
 6     2 service        0.0196
 7     2 south          0.0176
 8     2 invoices       0.0167
 9     2 departments    0.0157
10     2 days           0.0128
>
```

Figure 5: Top five words per topic with btea probabilities

Interpretation

- Topic 1 : Its top words are public (0.0239), government (0.0184), department (0.0157), coalition (0.0120) and administration (0.0102). Together they describe the structure and politics of government: public administration, departments and coalition arrangements. The word “public” leads the second word by a factor of about 1.3 (0.0239 / 0.0184), which makes it the clearest marker of this topic.
- Topic 2 : Its top words are service (0.0196), south (0.0176), invoices (0.0167), departments (0.0157) and days (0.0128). “Service”, “invoices”, “departments” and “days” fit the document 7 pattern, in which departments pay invoices within a number of days . “South” most likely comes from the place name “ South Africa”. The distribution within this topic is flatter than in Topic 1 : its strongest word leads the second by a factor of only about 1.1 (0.0196/0.0176).

2.3 Two complementary visualisation methods were used: a faceted bar chart that gives extract probability values and word clouds that show a larger set of words at a glance (Silge, '2017).

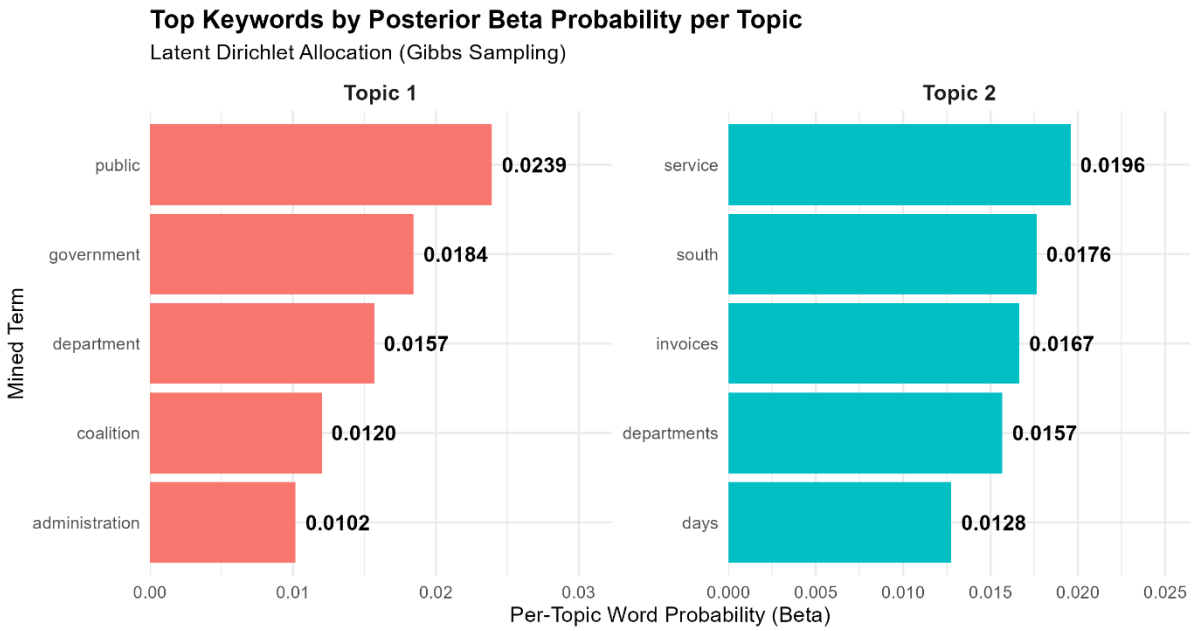


Figure 6: Top five keywords per topic by LDA beta probability

Interpretation

In topic 1, the bars fall steadily from “public” to “administration”, with “public” clearly the longest. In topic 2, the bars are closer in length, this shows the flatter distribution described above. Neither topic is dominated by one word.



Figure 7: Word clouds of the ten highest probability words

Interpretation

The clouds repeat the leading words of the bar chart and add five more words for each topic. Topic 1 adds management, governments, report, including and Africa, which stay within the vocabulary of public administration. Topic 2 adds national, eskom, police, psc and gxoyiya which are mostly names of public institutions or

organisations. Two words deserve attention: “including” and “report” are generic words with little topical value as well as “gxoyiya” which appears to be a person’s name that is repeated in an article. A custom stop word list could remove such terms. The split of “south” (Topic 2) and “africa” (Topic 1) also shows a limit of the bag of words model, which ignores word order and so cannot keep the phrase “South Africa” together.

Question 3

3.1 I chose the Our World in Data website as it is a public, non commercial research publication about global issues such as migration and mental health and its page consists of text rich descriptions on many topics, which make it suitable for a summarisation demonstration (Krotov, 2020).

The first three rows of webtext_df are shown below:

```
 3.1 SCRAPED DATA FRAME SAMPLE
> print(head(webtext_df, 3))
  paragraph_id
1            3
2            4
3            5

raw_text
1 How much of the population is international migrants, and how has this changed over time? Where do refugees
move from, and where do they go?
2                                                                                Hannah Ritchie, Esteban Ortiz-Ospina,
Tuna Acisu, and Simon van Teutem
3                         What is someone’s risk of developing depression, anxiety, or other mental health is
sues at some point in their life?
```

Figure 8: First three rows of the scraped data frame from the ' Our World in Data' website

Interpretation

The above shows a data frame with one row per paragraph. The identifiers start at 3, which shows that the first two paragraphs on the page were shorter than the length limit and were removed. Rows 1 and 3 are descriptive texts that pose questions about migration and mental health. Together they show that the home page covers many unrelated topics, so there is no single theme for a summary to capture. Row 2 is a list of authors. It passes the length filter although it is not content.

3.2 The cleaned text (clean_text) is used for counting and comparing. The original text (raw_text) is kept because a summary must show readable sentences with their capital letters and punctuation, not a stream of lower-case words (Anon., n.d.) (Denny, 2018).

3.3 Extractive methods were chosen as they do not need training data, they are easy to explain and keep the wording of the source , which suits a proof of concept. The two extractive techniques were applied: TextRank and frequency based sentence scoring (El-Kassas, 2021).

```
 QUESTION 3.3: EXTRACTIVE SUMMARIES
> cat("\n TextRank Graph-Based Summary \n"); print(summary_textrank)

 TextRank Graph-Based Summary
[1] "How much of the population is international migrants, and how has this changed over time?"
[2] "Where do refugees move from, and where do they go?"
[3] "Around one-fifth of development aid is spent on climate-related projects."
> cat("\n Frequency-Scored Summary \n"); print(summary_freq)

 Frequency-Scored Summary
[1] "The Spanish flu pandemic had a devastating impact on the global population."
[2] "What is someone's risk of developing depression, anxiety, or other mental health issues at some point in
their life?"
[3] "Explore updated data on vaccination coverage from the World Health Organization and UNICEF."
```

Figure 9: Extractive summaries (TextRank Graph Based Summary & Frequency- Scored Summary)

Interpretation: From the figure above both methods chose the same sentence about the long-run data on child mortality, which shows some overlap between TextRank and frequency scoring despite their different approaches. Together, the summaries cover a broad mix of topics from the page, including child mortality, global electricity from fossil fuels, renewables and nuclear since 1985, the description of Our World in Data as a project of Global Change Data Lab and the fact that one in four people in the world is younger than 15. Because the page itself is very broad, neither three-sentence summary can be fully representative, as each one only samples a few unrelated sections and leaves out most of the page's content. In terms of readability, the frequency-scored summary reads better because its sentences are complete, standalone facts, whereas the TextRank summary feels more disjointed. The poorest choice is the third TextRank sentence about Our World in Data being a project of Global Change Data Lab, as it is cut off at "(Reg." and is organisational boilerplate rather than informative page content. Overall, the Frequency Scored method produced the cleaner summary here, but both methods struggle to represent such a broad page in only three sentences.

- Visualisation 1: The bar chart shows the ten most frequent content words in the scraped text, which are the words that drive the frequency-based scores.

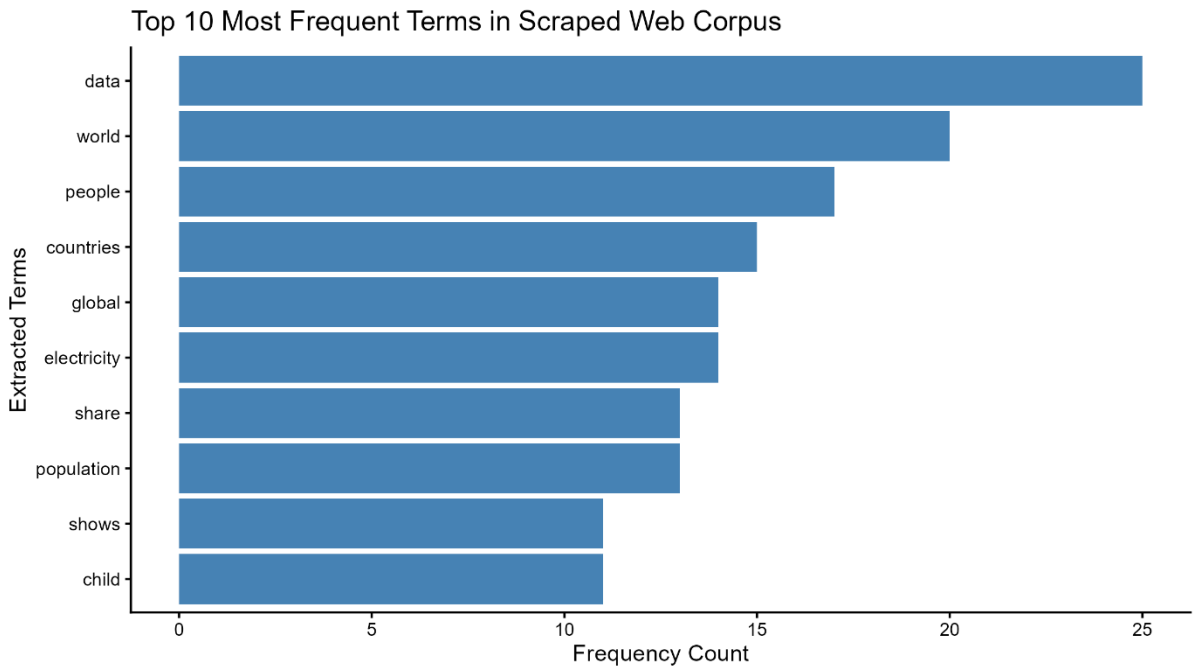


Figure 10: Ten most frequent terms in the scraped web corpus

Interpretation: The term “data” leads with a frequency of about 25, ahead of “world” with about 20 by 5 occurrences, followed by “people” with about 17 and “countries” with about 15. The next group is closely bunched, with “global” and “electricity” at about 14 each and “share” and “population” at about 13-14 each, while “shows” and “child” are lowest in the top 10 at about 11 each. These terms match the themes of the selected summary sentences well, as “data”, “world”, “countries”, “global” and “electricity” reflect the electricity and child mortality charts, and “people”, “population” and “child” reflect the demographic sentence about one in four people being younger than 15. However, the list is dominated by generic descriptive words such as “data”, “shows” and “share”, which come from repeated chart captions like “the chart shows” rather than from substantive content. No author names from bylines appear in the top 10, which suggests that byline or author-list terms were either removed during cleaning or were too infrequent to rank.

- Visualisation 2: The Figure below shows how many sentences fall into each length class (bins of 30 characters). The figure describes the units that the two summarisers choose from and so helps to interpret their output.

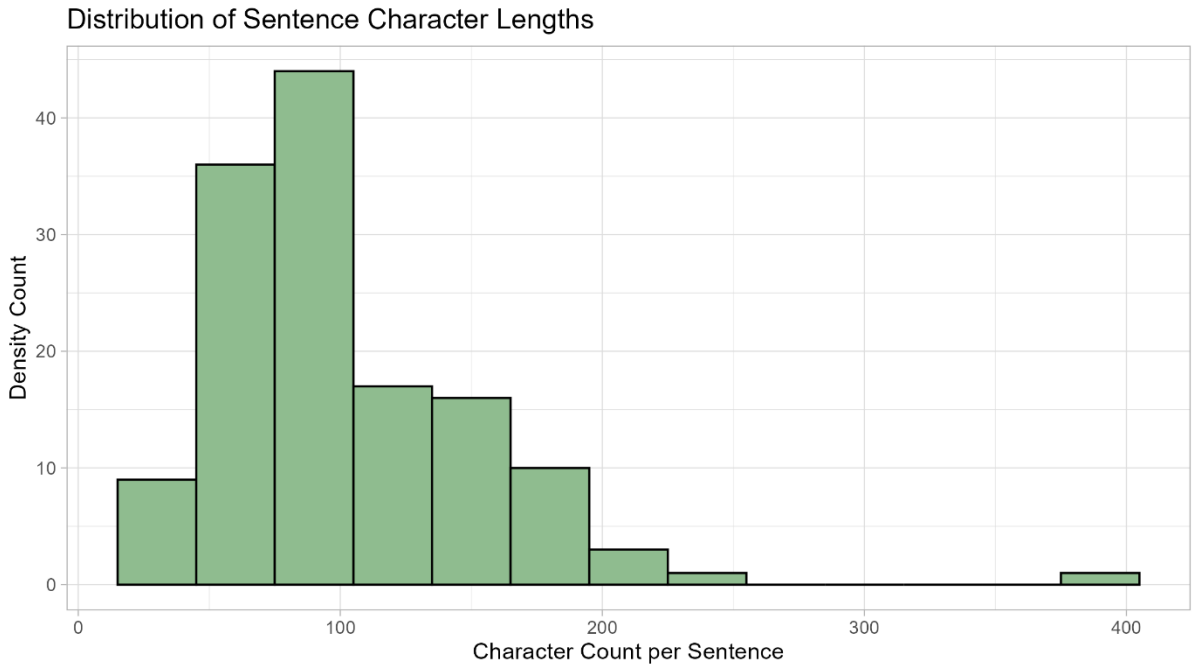


Figure 11: Distribution of sentence character lengths in the scraped text (the vertical axis shows the number of sentences)

Interpretation

The distribution is right-skewed with most sentences being short: the tallest bar is 75 to 105 characters with 44 sentences (about 32%), 80 of the 137 sentences (58%) are 45 to 105 characters and 106 sentences (77%) are shorter than 135 characters, or roughly 8 to 18 words, with the median in the 75 to 105 class, the mean likely slightly higher due to the long right tail, and only five sentences (4%) longer than 195 characters including one outlier of about 400 characters. This means the home page is made up of short, teaser-style sentences, so both summarisers choose from short units that each carry little information, the nine sentences below 45 characters are likely headings or fragments and are weak summary candidates, and the very long sentence is probably a sentence-splitting artefact where a paragraph without final punctuation, such as the author list, was merged with the next paragraph, which could be avoided by splitting each paragraph separately.

- Visualisation 3: The Figure below shows the ten highest-scoring sentences ranked by their average word frequency score. The figure shows how strongly the frequency method favours sentences and so helps to interpret why it selected the sentences in its summary.

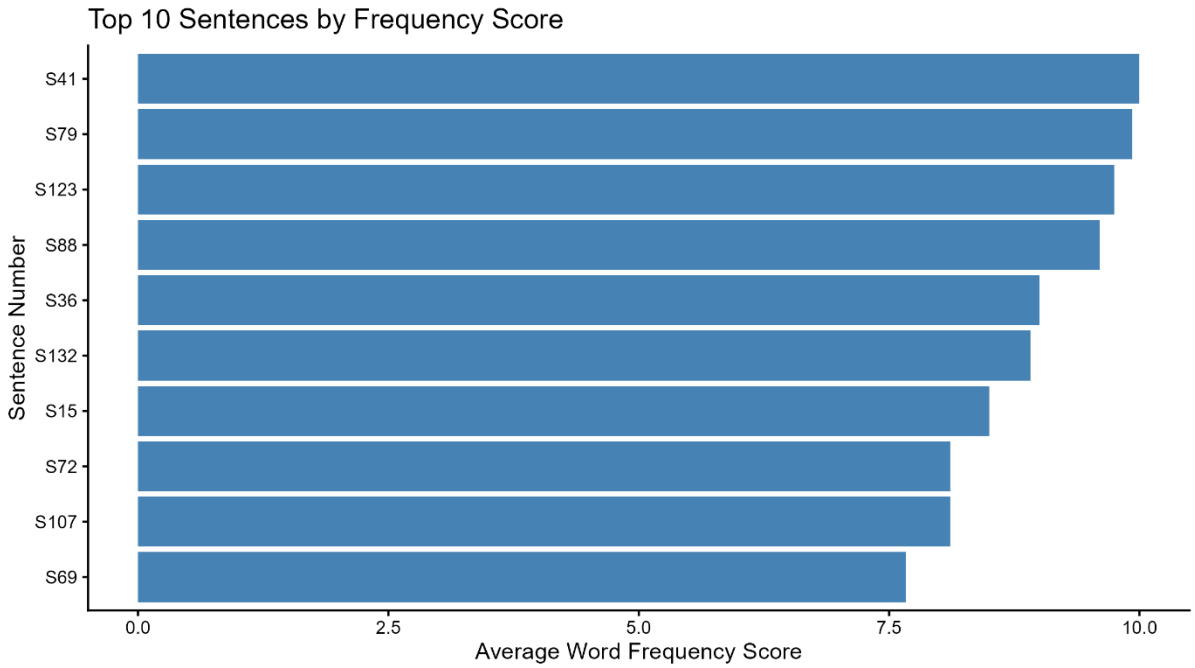


Figure 12: Top 10 Sentences by Average Word Frequency Score

Interpretation: The top 10 sentences have very similar scores, ranging only from about 7.7 for S69 to 10.0 for S41, a difference of just 2.3 points. The top four sentences: S41, S79, S123 and S88 are closely bunched between 9.4 and 10.0, while the remaining six fall gradually from 8.9 to 7.7. This narrow spread shows that the frequency method struggles to clearly separate the best sentences, because most sentences on the page reuse the same high-frequency words such as “data”, “world” and “people”. As a result, a high score here reflects the use of common words rather than necessarily more informative or representative content.

Question 4

4.1

Use Case

A research communications team wants to monitor the content that Our World in Data climate-change publishes on its climate, energy and environment topic pages. Every page is built from sections with a heading, a text and sometimes an author and date. The team needs one structured table in which every section has its title, text, author, publication date and an engagement measure (views), so that analysts can compare text length with engagement and find the best performing sections. The pages are unstructured HTML and publish no view counts.

The eight pages gave 114 raw records. After the removal of duplicates and page furniture, 76 unique records remained, which meets the requirement of atleast 20. Authors were found for 14 and dates for 13 of the 76 records. The text length of the 76 records is very uneven: the shortest text has 51 characters, the median is 232, the mean is 521 and the longest is 3316 characters. The mean is far above the

median, which means that a few article paragraphs are mixed with many short chart descriptions.

```
4.1 RAW NOISY DATASET
> print(head(raw_etl_df, 8))
  record_id
1         1
2         2
3         3
4         4
5         5
6         6
7         7
8         8
                                                                                  title_raw
1                                                 More people care about climate change than you think
2                                            How much have temperatures risen in countries across the world?
3                                 How much CO2 can the world emit while keeping warming below 1.5°C and 2°C?
4                                  Human greenhouse gas emissions have increased global average temperatures
```

```
R 4.6.1 · ~/ITBDA4-14 Project 2 - Amaka Nwaekwu/
5                    Global emissions have increased rapidly over the last 50 years and have not yet peaked
6 Current climate policies will reduce emissions, but not enough to keep temperature rise below 2°C
7                               <b>There are large differences in emissions across the world</b> &amp;amp; more
8                                Per capita, national, historical: how do countries compare on CO2 metrics?

content_raw
1
The majority of people in every country support action on climate, but the public consistently under
estimates this share.
2
Explore country-by-country data on monthly temperature anomalies.
3
The budget to keep temperatures below 1.5°C is less than a decade of current emissions. For 2°C, i
t's less than three decades.
4 Human emissions of carbon dioxide and other greenhouse gases are the primary drivers of the global
rise in temperatures.1 This link between global temperatures and greenhouse gas concentrations – esp
```

```
ecially CO2 – has been true throughout Earth's history.2 In the chart, we see the global average tem
perature relative to a baseline, which is the average between 1861 and 1890. This gives us an estima
te of the temperature increase compared to "pre-industrial" times. Average global temperatures have
risen by approximately 1.3°C since then (with even higher anomalies in the last few years). This war
ming has not been equally distributed across the world. The Northern Hemisphere has warmed more than
the Southern Hemisphere. In some regions, temperatures have risen by more than 5°C. You can see this
distribution in maps published by Berkeley Earth. Human emissions have been the main driver of this
change. Aerosols have played a slight cooling role in global climate, and natural variability has pl
ayed a minor role. This article from Carbon Brief explains this very well, with interactive graphics
showing the relative contributions of different factors to the climate.
5
6                                    Current policies to reduce, or at least slow down, the grow
th of CO2 emissions have already averted some future warming compared to a world without these polic
ies. As we see in the chart, climate and energy policies – which include adopted and implemented pol
icies (such as legislative decisions and executive orders) would reduce warming relative to a world
with no climate policies. This chart maps out future greenhouse gas emissions scenarios under a rang
```

```
Console  Terminal  Background Jobs
R 4.6.1 · ~/ITBDA4-14 Project 2 - Amaka Nwaekwu/
e of assumptions: if no climate policies were implemented; if current policies continued; if all cou
ntries achieved their current future pledges for emissions reductions; and the necessary pathways co
mpatible with limiting warming to 1.5°C or 2°C of warming this century.4 Current policies have us on
track for around 2.7°C by 2100. And if countries achieved their current pledges, this could be reduc
ed to 2.1°C. But if we aim to limit warming to "well below 2°C" – as is laid out in the Paris Agreem
ent – currently policies have us far off track. To achieve this goal, countries need to increase the
ambition of these commitments and bring their policies in line with them.
7
There is no single metric that captures the distribution of CO2 emissions across the world, today or
in the past. We can compare the total emissions of countries every year. But this fails to take popu
lation size into account; you'd expect a country with more people to have higher emissions. Even the
n, comparisons of total or per capita emissions today don't capture historical contributions. Some c
ountries that now have relatively low emissions have had high emissions for centuries. These metrics
tell us different things. We make all of them available by country in our data explorer. In the char
t, we see one metric to compare across the world: emissions per person. There are massive difference
s in emissions across the globe. Some of the wealthiest countries emit 100 times more per person tha
```

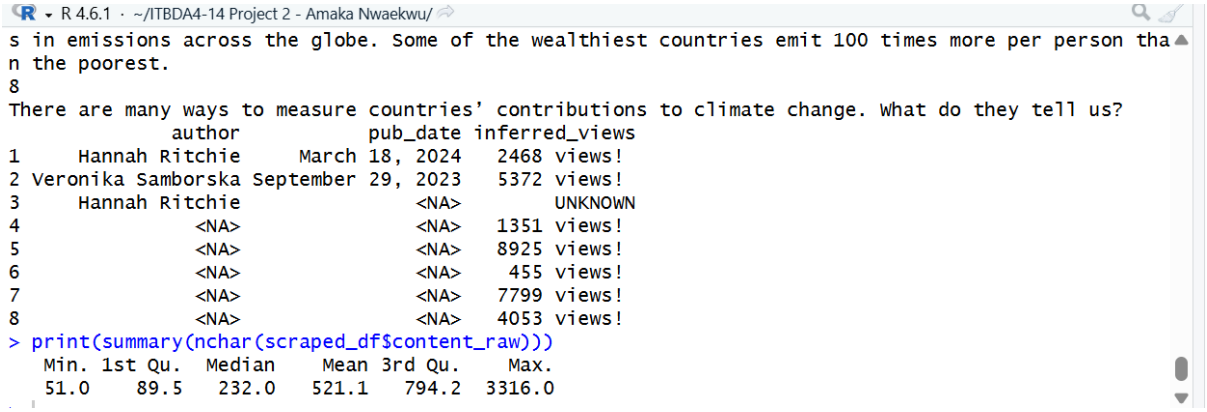


Figure 13: First eight records of the raw noisy dataset

Interpretation

The regular expressions did their work, where records 1 and 2 have their author and date in separate columns and the text no longer contains the byline. Record 3 has an author but no date. Its date probably stood under the heading “More articles on climate change”, which is page furniture and was removed together with its content, so a heading placed between an article and its date can separate information that belongs together. Records 4 to 8 have no author or date because they are sections without a byline, which is the natural incompleteness of web data. The injected noise is visible in record 3 (“UNKNOWN”), record 5 (empty text) and record 7 (markup in the title). The special characters and the untidy view counts cause errors if the data were analysed without being cleaned.

4.2 Only 2 of the 25 view counts were missing (8%), so imputation changes little. The median was chosen because it is not affected by extreme values. Replacing missing values by one central value keeps the centre of the data but reduces its variation and with a larger share of missing data, a method such as multiple imputation would be preferred (). Missing authors were set to “unknown” and missing dates were left empty because a date cannot be inferred from the other fields (Wickham, n.d.).

id	title_clean(start)	content_clean(start)	views	author	pub_date
1	more people care about climate change than you think	the majority of people in every country support action on climate but the public consistently underestimates this share	2468	Hannah Ritchie	2024-03-18
2	how much have temperatures risen in countries across the world	explore country by country data on monthly temperature anomalies	5372	Veronika Samborska	2023-09-29

3	how much co can the world emit while keeping warming below c and c	the budget to keep temperatures below c is less than a decade of current emissions for c it s less than three decades	5567	Hannah Ritchie	NA
4	human greenhouse gas emissions have increased global average temperatures	human emissions of carbon dioxide and other greenhouse gases are the primary drivers of the global rise in temperatures this link between global temperatures and greenhouse gas concentrations especially co has been true throughout earth s history in the chart we see the global average temperature relative to a baseline which is the average between and this gives us an estimate of the temperature increase compared to pre industrial times average global temperatures have risen by approximately c since then with even higher anomalies in the last few years this warming has not been equally distributed across the world the northern hemisphere has warmed more than the southern hemisphere in some regions temperatures have risen by more than c you can see	1351	unknown	NA

		this distribution in maps published by berkeley earth human emissions have been the main driver of this change aerosols have played a slight cooling role in global climate and natural variability has played a minor role this article from carbon brief explains this very well with interactive graphics showing the relative contributions of different factors to the climate			
5	global emissions have increased rapidly over the last years and have not yet peaked	No content available	8925	unknown	NA
6	current climate policies will reduce emissions but not enough to keep temperature rise below c	current policies to reduce or at least slow down the growth of co emissions have already averted some future warming compared to a world without these policies as we see in the chart climate and energy policies which include adopted and implemented policies such as legislative decisions and executive orders would reduce warming relative to a world with no climate policies this chart maps out future greenhouse gas emissions scenarios under a range of	455	unknown	NA

		assumptions if no climate policies were implemented if current policies continued if all countries achieved their current future pledges for emissions reductions and the necessary pathways compatible with limiting warming to c or c of warming this century current policies have us on track for around c by and if countries achieved their current pledges this could be reduced to c but if we aim to limit warming to well below c as is laid out in the paris agreement currently policies have us far off track to achieve this goal countries need to increase the ambition of these commitments and bring their policies in line with them			
7	there are large differences in emissions across the world more	there is no single metric that captures the distribution of co emissions across the world today or in the past we can compare the total emissions of countries every year but this fails to take population size into account you d expect a country with more people to have higher emissions even then comparisons of total or per capita	7799	unknown	NA

		emissions today don t capture historical contributions some countries that now have relatively low emissions have had high emissions for centuries these metrics tell us different things we make all of them available by country in our data explorer in the chart we see one metric to compare across the world emissions per person there are massive differences in emissions across the globe some of the wealthiest countries emit times more per person than the poorest			
8	per capita national historical how do countries compare on co metrics	there are many ways to measure countries contributions to climate change what do they tell us	4053	unknown	NA

Table 1: First eight records of the cleaned dataset

Interpretation

The text is now uniform, lower case letters and single spaces with no markup symbols or numbers and every view count is a whole number (record 3, which was “UNKNOWN” now has 5567). The empty content of record 5 carries the placeholder and the dates are actual real dates. The results also shows the limit of the cleaning. Removing the digits of “1.5˚C” leaves a single letter behind (“below c and c”) and apostrophes leave fragments such as “it s” and “you d”. The injected text “& more” in the title of record 7 is visible text, so the word “more” correctly remains once the tags and the entity are removed. Numbers inside the text are lost together with their meaning. Finally, authors and dates exist for only 6 of the 25 records, so these two columns are mostly incomplete.

4.3 The table web_analytics was created with six columns that match the cleaned CSV file and data/cleaned_web_data.csv was imported into it (Toman, 2021). The empty date fields of the CSV file are loaded as NULL. The table definition starts with

DROP TABLE IF EXISTS, so it must be run first then the import and only then the verification and analysis queries, running the whole script again would empty the table.

Table 2: Structure of the PostgreSQL table

Column	Type	Description
record_id	INTEGER (primary key)	Unique number of the scraped record
title_clean	TEXT	Cleaned title of the page section
content_clean	TEXT	Cleaned text of the section or “no content available”
numeric_views	INTEGER	Number of views with imputed values where it was missing
author	TEXT	Author names or “unknown”
pub_date	DATE	Publication date; NULL when unknown

Five queries check that the load is complete, that the imputation survived the import, how many authors and dates are missing and what the median text length (which Query 1 uses as its threshold) (Wickham, n.d.):

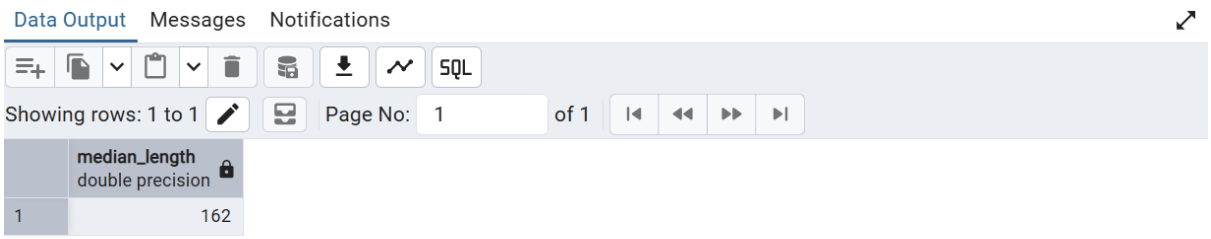
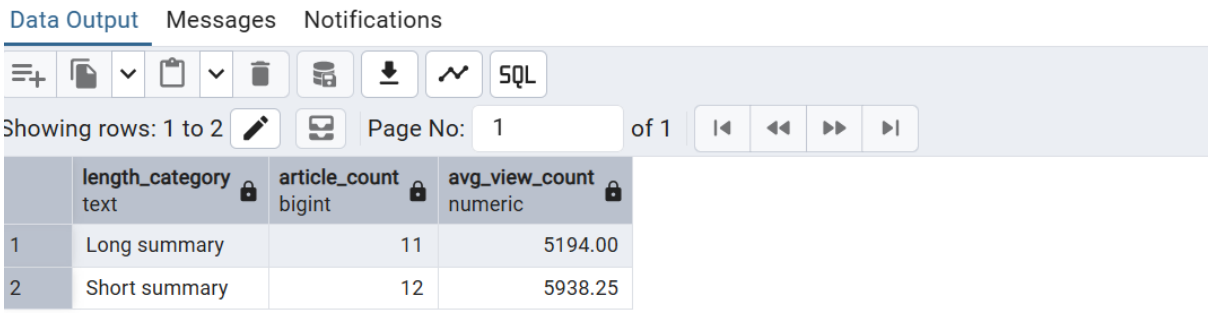


Figure 14:Median summary length



Data Output | Messages | Notifications

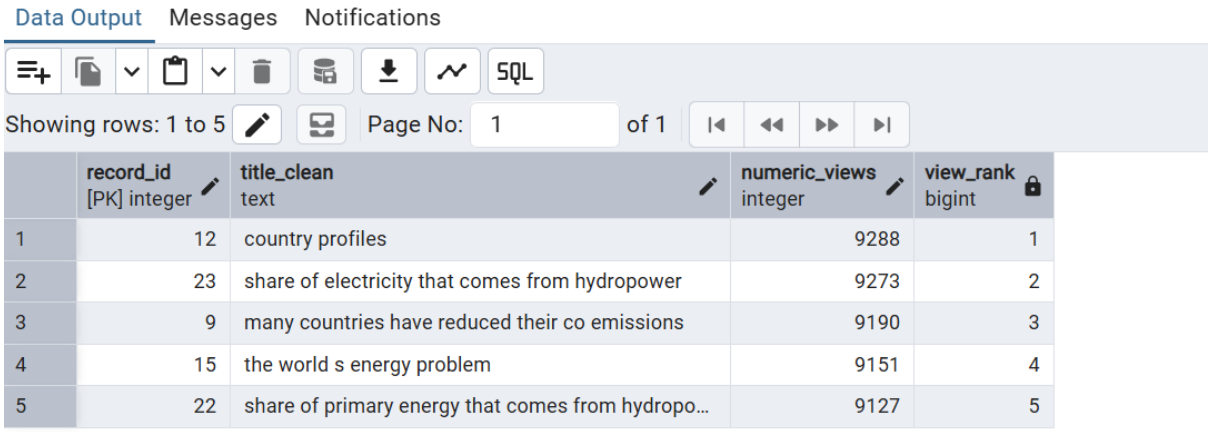
Showing rows: 1 to 2 | Page No: 1 of 1

	length_category text	article_count bigint	avg_view_count numeric
1	Long summary	11	5194.00
2	Short summary	12	5938.25

Figure 15: Query 1: Summary length versus engagement

Interpretation

Short summaries were more numerous (n = 12) and had a higher average view count of 5938.25 as compared to Long summaries with 5194 views. Since the views are random, the difference reflects chance. The query demonstrates the method but supports no conclusion about length and engagement.



Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	record_id [PK] integer	title_clean text	numeric_views integer	view_rank bigint
1	12	country profiles	9288	1
2	23	share of electricity that comes from hydropower	9273	2
3	9	many countries have reduced their co emissions	9190	3
4	15	the world s energy problem	9151	4
5	22	share of primary energy that comes from hydropo...	9127	5

Figure 16: Query 2: Top five articles by views

Table 3: The five records with the most views

record_id	**numeric_views**	**view_rank**	**subject of the stored title**
12	9288	1	"Country profiles" (reads like a navigation section)
23	9273	2	Share of electricity that comes from hydropower (a chart section)
9	9190	3	Many countries have reduced their CO_2 emissions
15	9151	4	The world's energy problem

22	9127	5	Share of primary energy that comes from hydropower (a chart section)

Interpretation

Record 12 has the most views (9,288). The five highest values lie within a band of only 161 views (9,288 to 9,127, about 1.7% of the leading value), so the ranking is decided by small differences. All five values are different, so the ranks 1 to 5 are consecutive and RANK and DENSE_RANK would give the same answer here (Fan, 2024). The subjects show what the table contains. "Country profiles" (record 12) reads like a navigation section that points to country pages, not like an article; records 23 and 22 are chart sections about hydropower; records 9 and 15 are article sections about CO_2 emissions and the world's energy problem. The table therefore mixes articles, chart sections and navigation sections and a more precise analysis would add a column for the type of section, as the view counts are simulated, the ranking shows how the query works and not which sections readers prefer.

References

I. Anon., n.d. s.l.: s.n.

II. Denny, M. a. S. A., 2018. ext preprocessing for unsupervised learning: Why it matters, when it misleads, and what to do about it. *Political analysis,* 26(2), pp. 168-189.

III. El-Kassas, W. S. C. R. A. a. M. H., 2021. Automatic text summarization: A comprehensive survey. *Expert systems with applications,* Volume 165, p. 113679.

IV. Fan, Y. H. Z. R. T. H. C. J. Y. Z. K. a. W. X., 2024. Metasql: A generate-then-rank framework for natural language to sql translation. *In 2024 IEEE 40th International Conference on Data Engineering,* pp. 1765-1778.

V. Ginde, G., Aedula, R., Saha, S., Mathur, A., Dey, S.R., Sampatrao, G.S. and Sagar, B.D., 2017. *Big Data Acquisition, Preparation, and Analysis Using Apache Software Foundation Tools*. New York: Chapman and Hall/CRC.

VI. Krotov, V. J. L. a. S. L., 2020. *Tutorial: Legality and ethics of web scraping,* s.l.: s.n.

VII. Mazumdar, S., Seybold, D., Kritikos, K. and Verginadis, Y., 2019. A survey on data storage and placement methodologies for cloud-big data ecosystem. *Journal of Big Data,* 6(1), pp. 1-37.

VIII. Silge, J. R. D. a. R. D., '2017. *Text mining with R: A tidy approach*. O'reilly ed. Boston: s.n.

IX. Toman, S., 2021. Web application for Data Import from XLSX into a Relational Database.

X. Wickham, H. a. G. G., n.d. R for data science. *Sebastopol: O'reily,* Volume 2.

XI. Zhang, Y. C. J. a. L. Q., 2018. MapReduce implementation of an improved XML keyword search algorithm. *Computer Systems ence and Engineering,* 33(2), pp. 125-135.